

Ecommerce Sales & Profitability Analysis

1. Project Overview

This project analyzes the sales performance and profitability of an e-commerce platform operating across 20 major Indian cities. By examining 5,000 transaction records from 2024 to 2025, the study aims to identify high-margin regions, evaluate the impact of discounting strategies, and understand consumer payment preferences to drive data-backed business growth.

2. Dataset Summary

- **Total Records:** 5,000 orders
- **Timeline:** 2024 – 2025
- **Geographic Scope:** 20 Cities across 4 Regions (North, South, East, West)
- **Product Range:** 10 Categories and 50 Sub-Categories (Books, Electronics, Groceries, etc.)
- **Key Features:**
 - **Order Details:** Order ID, Date, Product Name, Quantity
 - **Financials:** Sales, Unit Price, Discount (0–20%), Profit
 - **Customer Info:** Customer Name, City, Region
 - **Transaction:** Payment Mode (UPI, Credit Card, COD, etc.)
- **Data Quality:** The dataset is clean with zero null values in critical columns.

3. Exploratory Data Analysis (EDA) using Python

Initial exploration was conducted using Python to understand the distribution and statistical health of the data:

- **Data Loading:** Dataset imported using pandas.
- **Structural Check:** Performed `df.info()` to confirm data types (Dates as objects, Sales/Profit as floats).

```
df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 5000 entries, 0 to 4999
Data columns (total 14 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Order ID              5000 non-null  int64
1   Order Date            5000 non-null  datetime64[ns]
2   Customer Name         5000 non-null  object
3   Region                5000 non-null  object
4   City                  5000 non-null  object
5   Category              5000 non-null  object
6   Sub-Category          5000 non-null  object
7   Product Name          5000 non-null  object
8   Quantity              5000 non-null  int64
9   Unit Price            5000 non-null  int64
10  Discount              5000 non-null  int64
11  Sales                 5000 non-null  float64
12  Profit                5000 non-null  float64
13  Payment Mode          5000 non-null  object
dtypes: datetime64[ns](1), float64(2), int64(4), object(7)
memory usage: 547.0+ KB
```

- **Descriptive Statistics:** Used df.describe() to find

```
df.describe()
```

	Order ID	Order Date	Quantity	Unit Price	Discount	Sales	Profit
count	5000.000000	5000	5000.000000	5000.000000	5000.000000	5000.000000	5000.000000
mean	12500.500000	2024-09-30 18:58:27.840000	2.992600	39760.904600	10.051000	106733.204870	15941.746982
min	10001.000000	2023-10-04 00:00:00	1.000000	222.000000	0.000000	264.100000	19.120000
25%	11250.750000	2024-04-05 00:00:00	2.000000	20312.250000	5.000000	39766.537500	4892.295000
50%	12500.500000	2024-09-29 00:00:00	3.000000	39459.500000	10.000000	83080.325000	11108.525000
75%	13750.250000	2025-03-30 00:00:00	4.000000	59721.750000	15.000000	156968.587500	22467.987500
max	15000.000000	2025-10-03 00:00:00	5.000000	79998.000000	20.000000	398485.000000	89688.440000
std	1443.520003	NaN	1.413133	22831.783946	7.084662	85108.208202	14897.684916

- **Null Value Detection :**Used df.isnull() to find

```
df.isnull().sum()
```

```
Order ID      0
Order Date    0
Customer Name  0
Region        0
City          0
Category      0
Sub-Category  0
Product Name  0
Quantity      0
Unit Price    0
Discount      0
Sales         0
Profit        0
Payment Mode   0
dtype: int64
```

4. Advanced Data Analysis (SQL Queries)

To answer core business questions, the following SQL logic was applied:

1. **Top Profitable Cities:** Ranked cities by total profit to identify regional hubs.

city	Total_Profit
Lucknow	4523681
Guwahati	4487271
Bangalore	4416915
Chandigarh	4375380
Jaipur	4353578

2. **Discount Impact:** Calculated $\text{AVG}(\text{Profit}/\text{Sales})$ grouped by Discount level to see if high discounts eroded margins.

discount	Avg_Profit_Margin_Percent
0	14.9
5	14.95
10	14.96
15	15.03
20	14.8

3. **Revenue vs. Profit:** Compared total sales and profit margins across 10 categories.

category	Total_Sales	Profit_Margin_Percent
Home Decor	57233222	14.95
Furniture	56647188	15.35
Clothing	55053908	15.34
Books	54932643	14.7
Kitchen	54227902	14.53
Electronics	52587884	15.29
Toys	52227366	15.29
Sports	52069397	14.86
Beauty	50803410	14.2
Groceries	47883103	14.78

- Regional Demand:** Used RANK() OVER (PARTITION BY Region...) to find the top 3 sub-categories per region.

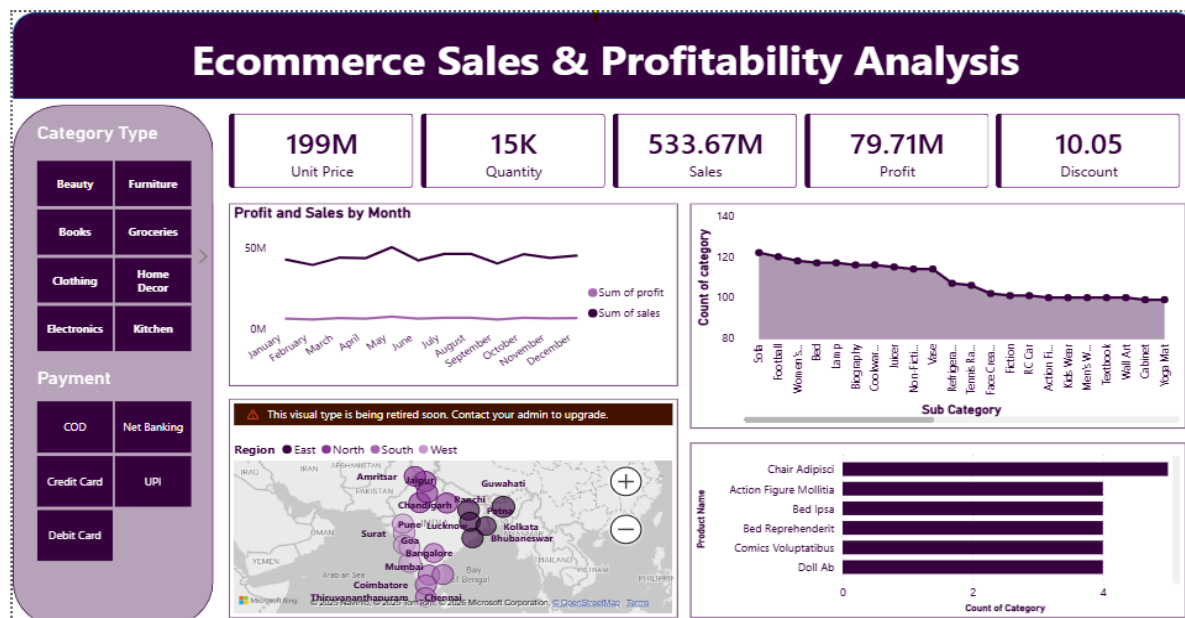
region	sub_category	Total_Quantity
East	Sofa	112
East	Oil	111
East	Football	104
North	Camera	112
North	Action Figure	109
North	Biography	107
South	Juicer	117
South	Lamp	102
South	Doll	102
West	Sofa	116
West	Microwave	112
West	Cookware Set	101

- Payment Efficiency:** Analyzed Average Order Value (AOV) across different payment modes.

payment_mode	Average_Order_Value
Net Banking	110361.9
COD	108339.7
Credit Card	106667.15
Debit Card	105031.29
UPI	103183.79

5. Dashboard In Power BI

Finally, I built an interactive dashboard in Power BI to present insights visually.



6. Business Recommendations

- **Regional Strategy:** Increase marketing spend in the **South** and **North** regions, which show the highest consistency in high-value orders.
- **Discount Optimization:** Limit 20% discounts to high-margin categories only, as data suggests margins stabilize better at 10-15% levels.
- **Inventory Allocation:** Focus stock of "Sofa" and "Rice" in the **North** region and "Laptops" and "T-shirts" in the **South** to meet localized demand.
- **Payment Incentives:** Promote **Credit Card** and **UPI** usage through small cashback offers, as these methods correlate with higher Average Order Values (AOV).